

Exp No: 12

EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

OBSERVATION:

- Two 16-bit binary numbers were multiplied using Embedded C.
- The multiplication produced a 32-bit result.
- The 32-bit result was displayed through **P0, P1, P2 and P3** of AT89C51.

PROGRAM:

```
#include <reg51.h> void main()
{
while (1)
{
unsigned int num1, num2; unsigned long int product; num1 = 0x2222;
num2 = 0BBBB;
product = (unsigned long int)num1 * num2; P0= product &0xff;
P1= (product & 0xff00)>>8;
P2 =(product & 0xff0000)>>16; P3 =(product & 0xff000000)>>24;
}
}
```

OUTPUT:

The screenshot displays the uVision IDE interface with the following components:

- Registers:** A list of registers (r0-r7, Sys, a, b, sp, sp_max, dptr, PC, status, sec, psw) and their current values.
- Disassembly:** A window showing the assembly code for the current instruction.
- main.c:** The source code file, showing a while loop that calculates a product and shifts its bits into four parallel ports (P0, P1, P2, P3).
- Parallel Port 0:** A window showing the value of P0 (0x06) and its bits (0, 1, 2, 3, 4, 5, 6, 7).
- Parallel Port 1:** A window showing the value of P1 (0xC4) and its bits (0, 1, 2, 3, 4, 5, 6, 7).
- Parallel Port 2:** A window showing the value of P2 (0x07) and its bits (0, 1, 2, 3, 4, 5, 6, 7).
- Parallel Port 3:** A window showing the value of P3 (0x19) and its bits (0, 1, 2, 3, 4, 5, 6, 7).
- Command:** A window showing the running status and the command line.
- Call Stack - Locals:** A window showing the current function call (MAIN) and its location/value (C:0x0800).